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Quantitative EEG (Brain Mapping)

Number: 0221

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Scope of Policy

This Clinical Policy Bulletin addresses quantitative EEG (brain mapping).

I. **Medical Necessity**

Aetna considers the use of quantitative EEG (brain mapping), also known as BEAM (Brain Electrical Activity Mapping), medically necessary only as an adjunct to traditional EEG for *any* of the following:

A. For ambulatory recording of EEG to facilitate subsequent expert visual EEG interpretation; *or*
B. For continuous EEG monitoring by frequency-trending to detect early, acute intracranial complications in the operating room or intensive care unit (ICU); *or*
C. For evaluation of certain members with symptoms of cerebrovascular disease whose neuroimaging and routine EEG studies are not conclusive; *or*
D. For evaluation of dementia and encephalopathy when the diagnosis remains unresolved after initial clinical evaluation; *or*
E. For screening for possible epileptic seizures in high-risk ICU members; *or*
F. For screening for possible epileptic spikes or seizures in long-term EEG monitoring; *or*
G. For topographic voltage and dipole analysis in pre-surgical evaluations for intractable epilepsy.

II. **Experimental, Investigational, or Unproven**

A. Aetna considers the use of quantitative EEG experimental, investigational, or unproven for all other indications, including any of the following diagnoses because there is inadequate scientific evidence to prove its clinical usefulness for these indications:

- Alcoholism
- Anxiety
- As a marker of central neuropathic pain intensity in spinal cord injury
- As a prognostic biomarker for long-term outcomes in preterm infants with neurological and medical complications
- Asperger syndrome and other autism spectrum disorders

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- Attention disorders (including attention deficit hyperactivity disorder)
- Bipolar disorder
- Chronic pain (diagnosis and guide to strategies for pain control)
- Depressed mood after stroke (screening)
- Depression (including monitoring and predicting the response of treatment)
- Drug abuse
- Eating disorder
- Fibromyalgia
- Hypoxic ischemic encephalopathy
- Insomnia
- Learning disability
- Management of individuals after chimeric antigen receptor (CAR)-T cell therapy
- Management of individuals in coma
- Mild or moderate head injury
- Minimally conscious state/persistent vegetative state
- Monitoring therapy responses and prognosticating neurological recovery of individuals with COVID-19
- Obsessive compulsive disorder
- Panic disorder
- Parkinson's disease (including use as a biomarker in non-invasive brain stimulation therapy in Parkinson's disease)
- Post-concussion syndrome
- Post-traumatic stress disorder (PTSD)
- Prediction of clinical impairment in stroke
- Prediction of outcomes in children following cardiac arrest
- Predicting response to psychotropic medication
- Prion diseases
- Schizophrenia
- Sepsis-associated encephalopathy prognosis
- Sports concussion (diagnosis and assessment of recovery)
- Tinnitus.

B. BrainScope One system (Ahead 300) is considered experimental, investigational, or unproven for evaluation of concussion / traumatic brain injury because of insufficient evidence.

III. Related Policies

- CPB 0480 - Tourette's Syndrome

Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
95961	Functional cortical and subcortical mapping by stimulation and/or recording of electrodes on brain surface, or of depth electrodes, to provoke seizures or identify vital brain structures; initial hour of attendance by a physician or other qualified health care professional.

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Code	Code Description
+ 95962	each additional hour of attendance by a physician or other qualified health care professional (List separately in addition to code for primary procedure)

Other CPT codes related to the CPB:

38225	Chimeric antigen receptor T-cell (CAR-T) therapy; harvesting of blood-derived T lymphocytes for development of genetically modified autologous CAR-T cells, per day
38226	preparation of blood-derived T lymphocytes for transportation (eg, cryopreservation, storage)
38227	receipt and preparation of CAR-T cells for administration
38228	CAR-T cell administration, autologous
95812 - 95830	Electroencephalography

HCPCS code covered if selection criteria are met:

S8040	Topographic brain mapping

HCPCS codes not covered for indications listed in the CPB:

BrainScope One system (Ahead 300) - no specific code:

Other HCPCS codes related to the CPB:

Q2041	Axicabtagene ciloleucel, up to 200 million autologous anti-cd19 car positive viable t cells, including leukapheresis and dose preparation procedures, per therapeutic dose
Q2042	Tisagenlecleucel, up to 600 million car-positive viable t cells, including leukapheresis and dose preparation procedures, per therapeutic dose

ICD-10 codes covered if selection criteria are met (not all-inclusive):

F02.80	Dementia in other diseases classified elsewhere, without behavioral disturbance
F02.A0, F02.B0, F02.C0, F02.C4	Dementia in other diseases classified elsewhere, with behavioral disturbance
F03.90 - F03.C4	Unspecified dementia
F06.1, F06.8	Psychotic disorders with hallucinations and other specified mental disorders due to known physiological conditions
G40.001 - G40.919	Epilepsy and recurrent seizures
G92.00 - G92.9	Toxic encephalopathy
G93.1	Anoxic brain damage, not elsewhere classified
G93.40 - G93.49	Encephalopathy, not elsewhere classified [not covered for assessing prognosis of sepsis-associated encephalopathy]

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Code	Code Description
G97.31 - G97.32	Intraoperative hemorrhage and hematoma of a nervous system organ or structure complicating a procedure
I65.01 - I69.998	Occlusion and stenosis of precerebral arteries, occlusion of cerebral arteries, transient cerebral ischemia, acute, but ill-defined cerebrovascular disease, other and ill-defined cerebrovascular disease, and late effects of cerebrovascular disease
I97.810 - I97.821	Other intraoperative and postprocedural cerebrovascular infarction during surgery
R56.1	Post traumatic seizures
R56.9	Unspecified convulsions
T56.0x1A - T56.0x1S	Toxic effect of lead and its compounds, accidental (unintentional)

ICD-10 codes not covered for indications listed in the CPB:

A81.00 - A81.9	Atypical virus infections of central nervous system [Prion diseases]
F07.81	Postconcussional syndrome
F10.121	Alcohol abuse with intoxication delirium
F10.14	Alcohol abuse with alcohol-induced mood disorder
F10.150 - F10.159	Alcohol abuse with alcohol-induced psychotic disorder
F10.180 - F10.19	Alcohol abuse with other alcohol-induced disorders
F10.221	Alcohol dependence with intoxication delirium
F10.230 - F10.24	Alcohol dependence with withdrawal and alcohol-induced mood disorder
F10.250 - F10.29	Alcohol dependence with alcohol-induced psychotic, persisting amnestic, persisting dementia and other alcohol-induced disorders
F10.920 - F10.99	Alcohol use, unspecified, with intoxication, alcohol-induced mood, psychotic, persisting amnestic, persisting dementia and other and unspecified alcohol-induced disorders
F11.10 - F19.99	Drug induced mental disorders
F20.0 - F20.9	Schizophrenia
F25.0 - F25.9	Schizoaffective disorders
F30.10 - F39	Mood [affective] disorders
F34.1	Dysthymic disorder
F41.0-F41.9	Generalized/mixed/other specified/unspecified anxiety disorders
F42.2 - F42.9	Obsessive-compulsive disorder
F43.10 - F43.12	Post-traumatic stress disorder
F50.00 - F50.9	Eating disorders
F51.01	Primary insomnia
F51.02	Adjustment insomnia
F51.03	Paradoxical insomnia
F51.09	Other insomnia not due to a substance or known physiological condition
F80.0 - F89	Pervasive and specific developmental disorders
F90.0 - F90.9	Attention-deficit hyperactivity disorders
G20.A1 - G20.C	Parkinson's disease
G47.00	Insomnia, unspecified
G89.21 - G89.29	Chronic pain not elsewhere classified
H93.11 - H93.93	Tinnitus
H93.A1 - H93.A9	Pulsatile tinnitus
I46.2 – I46.9	Cardiac arrest
I97.120 – I97.121	Postprocedural cardiac arrest

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Code	Code Description
J12.82	Pneumonia due to coronavirus disease 2019
M79.7	Fibromyalgia
P07.30 - P07.39	Preterm [premature] newborn
P11.5	Birth injury to spine and spinal cord
P29.81	Cardiac arrest of newborn
P91.60	Hypoxic-ischemic encephalopathy (HIE), unspecified
P91.819	Neonatal encephalopathy, unspecified
Q00.0-Q07.9	Congenital malformations of the nervous system
R40.20 - R40.2444	Coma
R40.2A	Nontraumatic coma due to underlying condition
R40.3	Persistent vegetative state
S06.0x0A - S06.9x9S	Intracranial injury [excluding those with skull fracture]
S09.10xA - S09.11xS S09.19xA - S09.19xS, S09.8xxA - S09.90xS	Head injury, unspecified
S14.101A- S14.159S, S24.101A- S24.159S, S34.101A- S34.139S	Spinal cord injury
R48.0	Dyslexia and alexia
U07.1	COVID-19
W21.00xA - W21.9xxS	Striking against or struck by sports equipment
Z13.31	Encounter for screening for depression
Z73.810 - Z73.819	Behavioral insomnia of childhood
Z86.73	Personal history of transient ischemic attack (TIA), and cerebral infarction without residual deficits [depressed mood after stroke]
Z86.74	Personal history of sudden cardiac arrest

Background

Quantitative EEG (qEEG) is a method of analyzing the electrical activity of the brain to derive quantitative patterns that may correspond to diagnostic information and/or cognitive deficits.

Quantitative EEG, a technique for topographic display and analysis of brain electrophysiological data, has been proposed for use in the diagnosis of various psychiatric disorders. Clinical studies have demonstrated distinctive forms of brain electrical activity in psychiatric conditions including attention deficit disorder, schizophrenia, major depression, and obsessive-compulsive disorder. However, the clinical significance of

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these distinctive patterns of brain wave activity is unknown. Thus the role of quantitative EEG in diagnosis, evaluation of disease progression, and treatment of these conditions has yet to be elucidated. A report from the American Academy of Neurology and the American Clinical Neurophysiology Society concluded that quantitative EEG remains investigational for clinical use in post-concussion syndrome, mild-to-moderate head injury, learning disability, attention disorders, schizophrenia, depression, alcoholism, and drug abuse.

Clinical studies have demonstrated distinctive forms of brain electrical activity in neurologic and psychiatric conditions including learning disabilities, autism, traumatic brain injury, coma, schizophrenia, major depression, and obsessive-compulsive disorder. However, the clinical significance of these distinctive patterns of brain wave activity is unknown. Thus the role of quantitative EEG in diagnosis, evaluation of disease progression, and treatment of these conditions has yet to be elucidated.

Quantitative EEG has been proposed for use in a broad array of potential applications. This evidence has focused on the diagnostic accuracy of QEEG. There is, however, a paucity of evidence regarding its clinical utility.

There are no current guidelines from leading medical professional organizations recommending the use of quantitative EEG as a screening test for neurological and psychiatric conditions. In addition, there are no peer-reviewed published prospective studies of the use of quantitative EEG screening for these conditions showing that management is altered such that clinical outcomes are improved.

There are no published clinical studies demonstrating that use of quantitative EEG reduces the number of imaging studies or other follow-up tests. In addition, there are no current guidelines from leading medical professional organizations recommending the use of quantitative EEG either as a prerequisite to, or as a replacement for, imaging studies.

While there is some evidence that electroencephalograph activity differs between normal control subjects and subjects suffering from tinnitus, additional evidence is needed to evaluate the value of including quantitative EEG in a battery of electrophysiological tests for the clinical identification of a predominantly central type of tinnitus. In addition, there is little evidence to support the use of quantitative EEG to determine the need for change of medications in the treatment of tinnitus.

Some investigators have proposed use of quantitative EEG in psychiatric cases to facilitate selection of medications. However, there is a lack of reliable evidence from prospective studies demonstrating that clinical outcomes are improved by basing selection of psychotropic medications on quantitative EEG results compared to empiric selection. The FDA approved prescribing information for psychotropic medications includes no recommendation for use of quantitative EEG in selection or dosing, and there are no current guidelines from leading medical professional organizations recommending such use of quantitative EEG.

Crumbley and associates (2005) examined the use of quantitative EEG in predicting response to psychotropic medication. The clinical outcomes of 2 groups of patients were compared:

I. those with prescribed medication regimens that were concordant with the quantitative EEG predictors, and
II. those whose medication regimens were discordant with the quantitative EEG predictors.

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Participants included 70 adolescent inpatients who were administered quantitative EEG upon admission. The results indicated no significant difference in clinical outcome between the two groups. The failure of this study to find significant differences in patient outcomes questions this particular use of the quantitative EEG (Crumbley et al, 2005).

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John and Prichep (2006) noted that as quantitative EEG and pharmaco-EEG have evolved, a vast body of facts has been accumulated, describing changes in the EEG or event-related potentials observed in a variety of brain disorders or after administration of a variety of medications. With some notable exceptions, these studies have tended to be phenomenological rather than analytical. There has not been a systematic attempt to integrate these phenomena to provide better understanding of how the abnormal behaviors of a particular psychiatric patient might be related to the specific pattern of the deviant electrical activity, nor just how pharmacological reduction of that deviant activity may have resulted in more normal behavior.

There is insufficient evidence to support the use of quantitative EEG in the diagnosis and/or classification of attention-deficit hyperactivity disorder (ADHD) (Krull, 2009). Several studies have demonstrated differences in qEEG between groups of children with ADHD and normal children. However, these studies are limited by non-random assignment, lack of blinding, failure to consider comorbidities, and/or failure to control for pharmacologic therapy. In addition, the specificity of the findings for ADHD has not been demonstrated.

Snyder and Hall (2006) performed a meta-analysis on the use of quantitative EEG in evaluating patients with ADHD. The 9 eligible studies (n = 1,498) observed quantitative EEG traits of a theta power increase and a beta power decrease, summarized in the theta/beta ratio with a pooled effect size of 3.08 (95% confidence interval: 2.90 to 3.26) for ADHD versus controls (normal children, adolescents, and adults). These investigators concluded that this meta-analysis supports that a theta/beta ratio increase is a commonly observed trait in patients with ADHD relative to normal controls. Moreover, they noted that since it is known that the theta/beta ratio trait may arise with other conditions, a prospective study covering differential diagnosis would be needed to determine generalizability to clinical applications. Furthermore, standardization of the quantitative EEG technique is also needed, specifically with control of mental state, drowsiness, and medication.

Although QEEG may prove to be helpful in the diagnosis and/or classification of ADHD in the future, at present, there is insufficient evidence to support its use in clinical populations.

Much of the literature submitted focuses on the use of QEEG in the early detection of dementia. Although several markers of early dementia have been reported in the literature, there is a lack of evidence that early detection of dementia alters clinical management such that outcomes are improved, especially given the lack of robust treatments available.

An assessment by the Swedish Office of Health Technology Assessment (SBU, 2008) found insufficient evidence to support the use of quantitative EEG in dementia. The SBU assessment stated: "[t]here is limited evidence that either visually rated EEG or qEEG helps the diagnostic workup differentiate AD (Alzheimer's Disease) patients from controls or AD from other dementia disorders."

Klassen et al (2011) evaluated qEEG measures as predictive biomarkers for the development of dementia in Parkinson disease (PD). Preliminary work shows that qEEG measures correlate with current PD cognitive state. A reliable predictive qEEG biomarker for PD dementia (PD-D) incidence would be valuable for studying PD-D, including treatment trials aimed at preventing cognitive decline in PD. A cohort of subjects with PD in the authors' brain donation program utilizes annual pre-mortem longitudinal movement and cognitive evaluation. These subjects also undergo biennial EEG recording. EEG from subjects with PD without dementia with follow-up cognitive evaluation was analyzed for qEEG measures of background rhythm frequency and relative power in δ, α, and β bands. The relationship between the time to onset of dementia and qEEG and other possible predictors was assessed by using Cox regression. The hazard of developing dementia was 13 times higher for those with low background rhythm frequency (lower than the grand median of 8.5 Hz) than for those with high background rhythm frequency ($p < 0.001$). Hazard ratios (HRs) were also significant for greater than median bandpower (HR = 3.0; $p = 0.004$) compared to below, and for certain neuropsychological measures. The HRs for δ, α, and β bandpower as well as baseline demographic and clinical characteristics were not significant. The authors concluded that qEEG measures of background rhythm frequency and relative power in the band are potential predictive biomarkers for dementia incidence in PD. These QEEG biomarkers may be useful in complementing neuropsychological testing for studying PD-D incidence.

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Marzano and colleagues (2008) stated that in the last 2 decades quantitative EEG analysis has been used to examine the neurophysiological characteristics of insomnia. These studies provided evidence in support of the hypothesis that primary insomnia is associated with hyper-arousal of central nervous system and altered sleep homeostasis. However, these researchers have here underlined that these results have intrinsic methodological problems, mainly related to constraints of standard assessment in clinical research. They have proposed that future studies should be performed on larger samples of drug-free patients, using within-subjects designs and longitudinally recording patients adapted to sleep laboratory. All these methodological improvements will allow to partial out the contribution of individual differences, pharmacological influences and first-night effects on EEG frequencies. Moreover, they have discussed the potential relevance of recent findings from basic research concerning local changes during physiological sleep, which could be extended to the study of insomnia.

Hargrove and colleagues (2010) stated that there is increasing acceptance that pain in fibromyalgia (FM) is a result of dysfunctional sensory processing in the spinal cord and brain, and a number of recent imaging studies have demonstrated abnormal central mechanisms. These researchers compared quantitative electroencephalogram (qEEG) measures in 85 FM patients with age- and gender-matched controls in a normative database. A statistically significant sample (minimum 60 seconds from each subject) of artifact-free EEG data exhibiting a minimum split-half reliability ratio of 0.95 and test-retest reliability ratio of 0.90 was used as the threshold for acceptable data inclusion. Electroencephalograms of FM subject were compared to EEGs of age- and gender-matched healthy subjects in the Lifespan Normative Database and analyzed using NeuroGuide 2.0 software. Analyses were based on spectral absolute power, relative power and coherence. Clinical evaluations included the Fibromyalgia Impact Questionnaire (FIQ), Beck Depression Inventory and Fischer dolorimetry for pain pressure thresholds. Based on Z-statistic findings, the EEGs from FM subjects differed from matched controls in the normative database in 3 features:

I. reduced EEG spectral absolute power in the frontal International 10-20 EEG measurement sites, particularly in the low- to mid-frequency EEG spectral segments;
II. elevated spectral relative power of high frequency components in frontal/central EEG measurement sites; and
III. widespread hypo-coherence, particularly in low- to mid-frequency EEG spectral segments, in the frontal EEG measurement sites.

I. reduced EEG spectral absolute power in the frontal International 10-20 EEG measurement sites, particularly in the low- to mid-frequency EEG spectral segments;
II. elevated spectral relative power of high frequency components in frontal/central EEG measurement sites; and
III. widespread hypo-coherence, particularly in low- to mid-frequency EEG spectral segments, in the frontal EEG measurement sites.

A consistent and significant negative correlation was found between pain severity and the magnitude of the EEG abnormalities. No relationship between EEG findings and medicine use was found. The authors concluded that qEEG analysis reveals significant differences between FM patients compared to age- and gender-matched healthy controls in a normative database, and has the potential to be a clinically useful tool for assessing brain function in FM patients.

Hathi et al (2010) assessed an EEG-based index, the Cerebral Health Index in babies (CHI/b), for identification of neonates with high Sarnat scores and abnormal EEG as markers of hypoxic ischemic encephalopathy (HIE) after perinatal asphyxia. This was a retrospective study using 30-min EEG data collected from 20 term neonates with HIE and 20 neurologically normal neonates. The HIE diagnosis was made on clinical grounds based on history and examination findings. The maximum-modified clinical Sarnat score was used to grade HIE severity within 72 hrs of life. All neonates underwent 2-channel bedside EEG monitoring. A trained electroencephalographer blinded to clinical data visually classified each EEG as normal, mild or severely abnormal. The CHI/b was trained using data from Channel 1 and tested on Channel 2. The CHI/b distinguished among HIE and controls ($p < 0.02$) and among the 3 visually interpreted EEG categories ($p < 0.0002$). It showed a sensitivity of 82.4% and specificity of 100% in detecting high grades of neonatal encephalopathy (Sarnat 2 and 3), with an area under the receiver operator characteristic (ROC)

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curve of 0.912. CHI/b also identified differences between normal versus mildly abnormal ($p < 0.005$), mild versus severely abnormal ($p < 0.01$) and normal versus severe ($p < 0.002$) EEG groups. An ROC curve analysis showed that the optimal ability of CHI/b to discriminate poor outcome was 89.7% (sensitivity: 87.5%; specificity: 82.4%). The authors concluded that the CHI/b identified neonates with high Sarnat scores and abnormal EEG. These results support its potential as an objective indicator of neurological injury in infants with HIE.

Lopes et al (2010) examined and compared the brain cortical activity, as indexed by qEEG power, coherence and asymmetry measures, in panic disorder patients during an induced panic attack with a 35% CO(2) challenge test and also in a resting condition. A total of 15 subjects with panic disorder were randomly assigned to both 35% CO(2) mixture and atmospheric compressed air, in a double-blind study design, with EEG being recorded for a 20-min period. During induced panic attacks, a reduced right-sided frontal orbital asymmetry in the beta band, a decreased occipital frontal intra-hemispheric coherence in the delta band at both right and left sides, a left-sided occipital delta inter-hemispheric asymmetry and an increased relative power in the beta wave at T4 were observed. These data showed a disturbed frontal cortical processing, pointing to an imbalance of the frontal and occipital sites, common to both hemispheres, and an increased right posterior activity related to the high arousing panic attack condition. These findings corroborated the neuroanatomical hypothesis of panic disorder.

Velasques et al (2013) examined the relationship between cortical gamma coherence within patients with bipolar disorder and a control group during a pro-saccadic attention task. These investigators hypothesized that gamma coherence oscillations act as a main neural mechanism underlying information processing which changes in bipolar patients. A total of 32 subjects (12 healthy controls and 20 bipolar patients) were enrolled in this study. Participants performed a pro-saccadic attention task while their brain activity pattern was recorded using qEEG (20 channels). These researchers observed that the maniac group presented lower saccade latency when compared to depression and control groups. The main finding was a greater gamma coherence for control group in the right hemisphere of both frontal and motor cortices caused by the execution of a pro-saccadic attention task. The authors concluded that these findings suggested a disrupted connection of the brain's entire functioning of maniac patients and represented a deregulation in cortical inhibitory mechanism. Thus, these results reinforce the hypothesis that greater gamma coherence in the right and left frontal cortices for the maniac group produces a "noise" during information processing and highlights that gamma coherence might be a biomarker for cognitive dysfunction during the manic state. The authors stated that these findings need to be confirmed in larger samples and in bipolar patients before start the pharmacological treatment.

An UpToDate review on “Attention deficit hyperactivity disorder in children and adolescents: Clinical features and evaluation” (Krull, 2013) states that “We do not suggest qEEG for the evaluation of children with ADHD. Although the US Food and Drug Administration has licensed the first EEG test for assessment of children (6 to 17 years) for ADHD, and several studies have demonstrated differences in qEEG between children with ADHD and normal children, the studies were limited by non-random assignment, lack of blinding, failure to consider comorbidities, and/or failure to control for pharmacologic therapy. In addition, the EEG patterns differ in boys and girls. A 2013 meta-analysis of nine studies (including 1253 children with ADHD and 517 without ADHD) found significant heterogeneity and concluded that EEG profiles (specifically an increased theta to beta ratio) cannot be used to reliably diagnose ADHD (although they may be helpful for prognosis). Current evidence is insufficient to support the use of qEEG over clinical evaluation of symptoms and functional impairment for the diagnosis of ADHD”.

Kutcher et al (2013) summarized the evidence for the following technologies/strategies related to diagnosing or managing sports-related concussion: quantitative EEG, functional neuroimaging, head impact sensors, telemedicine and mobile devices. Databases used were MEDLINE, PubMed, Cochrane Controlled Trials Registers, SportDiscus, EMBASE, Web of Science and ProQuest databases. Primary search keywords were concussion, sports concussion and mild traumatic brain injury. The keywords used for secondary, topic specific searches were quantitative electroencephalography, qEEG, functional MRI, magnetoencephalography, near-infrared spectroscopy, positron emission tomography, single photon emission CT, accelerometer, impact sensor, telemetry, remote monitoring, robotic medicine, telemedicine, mobile device, mobile phone, smart phone and tablet computer. The primary search produced 8,567 publications.

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The secondary searches produced 9 publications that presented original data, included a comparison group in the study design and involved sports-related concussion: 4 studies spoke to the potential of qEEG as a diagnostic or management tool, while 5 studies addressed the potential of fMRI to be used in the same capacity. The authors concluded that emerging technologies and novel approaches that aid in sports concussion diagnosis and management are being introduced at a rapid rate. Moreover, they stated that while some technologies show promise, their clinical utility remains to be established.

Furthermore, the American Medical Society for Sports Medicine's position statement on "Concussion in sport" (Harmon et al, 2013) did not mention the use of quantitative EEG/brain mapping as a management tool.

Hosokawa et al (2014) noted that several studies have reported the presence of EEG abnormalities or altered evoked potentials (EPs) during sepsis. However, the role of these tests in the diagnosis and prognostic assessment of sepsis-associated encephalopathy remains unclear. These researchers performed a systematic search for studies evaluating EEG and/or EPs in adult patients with sepsis-associated encephalopathy. The following outcomes were extracted:

I. incidence of EEG/EP abnormalities;
II. diagnosis of sepsis-associated delirium or encephalopathy with EEG/EP; and
III. outcome.

I. incidence of EEG/EP abnormalities;
II. diagnosis of sepsis-associated delirium or encephalopathy with EEG/EP; and
III. outcome.

Among 1,976 citations, 17 articles met the inclusion criteria. The incidence of EEG abnormalities during sepsis ranged from 12% to 100% for background abnormality and 6% to 12% for presence of tri-phasic waves. Two studies found that epileptiform discharges and electrographic seizures were more common in critically ill patients with than without sepsis. In 1 study, EEG background abnormalities were related to the presence and the severity of encephalopathy. Background slowing or suppression and the presence of tri-phasic waves were also associated with higher mortality. A few studies demonstrated that quantitative EEG analysis and EP could show significant differences in patients with sepsis compared to controls; but their association with encephalopathy and outcome was not evaluated. The authors concluded that abnormalities in EEG and EPs are present in the majority of septic patients. They stated that there is some evidence to support EEG use in the detection and prognostication of sepsis-associated encephalopathy, but further clinical investigation is needed to confirm this suggestion.

Minimally Conscious State/Persistent Vegetative State

In a systematic review and meta-analysis, Bender et al (2015) examined the sensitivity and specificity of new diagnostic methods for the minimally conscious state (MCS). These researchers identified and evaluated 20 clinical studies involving a total of 906 patients with either persistent vegetative state (PVS) or MCS. The reported sensitivities and specificities of the various techniques used to diagnose MCS vary widely. The sensitivity and specificity of functional MRI-based techniques were 44% and 67%, respectively (with corresponding 95% confidence intervals [CI]: 19% to 72% and 55% to 77%); those of quantitative EEG were 90% and 80%, respectively (95% CI: 69% to 97% and 66% to 90%); EEG, event-related potentials, and imaging studies could also aid in prognostication. Contrary to prior assumptions, 10% to 24% of patients in PVS could regain consciousness, sometimes years after the event, but only with marked functional impairment. The authors concluded that the basic diagnostic evaluation for differentiating PVS from MCS consists of a standardized clinical examination. They stated that in the future, modern diagnostic techniques may help identify patients who are in a subclinical MCS.

Furthermore, an UpToDate review on "Hypoxic-ischemic brain injury: Evaluation and prognosis" (Weinhouse and Young, 2016) states that "The clinical value of the electroencephalogram (EEG) is unclear in the assessment of prognosis of anoxic brain injury because investigators have used different classification systems and variable intervals of recordings after resuscitation. Furthermore, the EEG is susceptible to

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subjective interpretation, the effects of sedative drugs, metabolic disturbances, and sepsis, which can invalidate the results".

Attention Deficit Hyperactivity Disorder (ADHD)

The American Academy of Neurology (AAN)'s practice advisory on "The utility of EEG theta/beta power ratio in ADHD diagnosis" (Gloss et al, 2016) evaluated the evidence for EEG theta/beta power ratio for diagnosing, or helping to diagnose ADHD. The authors identified relevant studies and classified them using AAN criteria. Two Class I studies assessing the ability of EEG theta/beta power ratio and EEG frontal beta power to identify patients with ADHD correctly identified 166 of 185 participants. Both studies evaluated theta/beta power ratio and frontal beta power in suspected ADHD or in syndromes typically included in an ADHD differential diagnosis. A bivariate model combining the diagnostic studies showed that the combination of EEG frontal beta power and theta/beta power ratio has relatively high sensitivity and specificity but is insufficiently accurate. The authors concluded that it is unknown whether a combination of standard clinical examination and EEG theta/beta power ratio increases diagnostic certainty of ADHD compared with clinical examination alone. The AAN provided the following recommendations:

Clinicians should inform patients with suspected ADHD and their families that the combination of EEG theta/beta power ratio and frontal beta power should not replace a standard clinical evaluation. There is a risk for significant harm to patients from ADHD misdiagnosis because of the unacceptably high false-positive diagnostic rate of EEG theta/beta power ratio and frontal beta power. Level B (Probably effective, ineffective or harmful (or probably useful/predictive or not useful/predictive) for the given condition in the specified population)

Clinicians should inform patients with suspected ADHD and their families that the EEG theta/beta power ratio should not be used to confirm an ADHD diagnosis or to support further testing after a clinical evaluation, unless such diagnostic assessments occur in a research setting. Level R (Level R recommendations are ones that "the guideline authors assert should be applied only in research settings).

Byeon and colleagues (2020) examined qEEG subtypes as auxiliary tools to evaluate ADHD. A total of 74 subjects (58 male and 16 female) were examined using the Korean version of the Diagnostic Interview Schedule for Children Version IV and were assigned to 1 of 3 groups: ADHD (n = 27), ADHD-not otherwise specified (NOS; n= 32), and neurotypical (NT; n = 15). These researchers measured absolute and relative EEG power in 19 channels and conducted an auditory continuous performance test. They analyzed qEEG according to the frequency range: delta (1 to 4 Hz), theta (4 to 8 Hz), slow alpha (8 to 10 Hz), fast alpha (10 to 13.5 Hz), and beta (13.5 to 30 Hz). Subjects were then grouped by Ward's method of cluster analysis using the squared Euclidian distance to measure dissimilarities. These investigators found 4 qEEG clusters, which were characterized by the following: First, elevated delta power with less theta activity. Second, elevated slow alpha relative power. Third, elevated theta with deficiencies of alpha. Fourth, beta relative power; and elevated fast alpha and beta absolute power. The largest proportion of participants in first and third clusters were from the ADHD group (48% and 47%, respectively). Conversely, the second group mostly consisted of subjects from the NOS group (59%), while the fourth group had the largest proportion of subjects from the NT group (62%). The authors stated that these findings indicated that children with ADHD did not neurophysiologically constitute a homogenous group. They also identified a new subtype with increased alpha power in addition to those commonly reported in ADHD. These investigators noted that given the qEEG characteristics with increased alpha power, clinicians should consider the possibility that this subtype may be caused by childhood depression. The authors believed that these qEEG subtypes of ADHD are expected to provide valuable information for accurately diagnosing ADHD.

The authors stated that this study had several drawbacks. First, these researchers failed to fully consider the IQ of the subjects, although they excluded subjects below an IQ of 70. It is known that EEG can vary depending on an individual's IQ, so this variable should be controlled for. Second, the number of subjects among the 3 groups (ADHD, NOS, NT) was inconsistent, especially the number of NT. Third, the study may have favored those more distracted and careless than ordinary children, because it targeted children who wanted to participate in ADHD research through posters. Fourth, although ADHD, NOS and NT each accounted for a major portion of all 4 groups, the proportion was still low because of the heterogenicity. A

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low proportion in the classification reduced the typicality and reliability of the findings; thus, the distinction between ADHD, NOS and NT group was limited by the characteristics of qEEG. However, from the perspective that the NOS group also needed therapeutic intervention, the significance of this study depended on revealing the characteristics of qEEG in the ADHD group, including the NOS group, and its differences from the NT group. Lastly, these investigators focused on subjects' attention and chose not to focus on other symptoms, such as depression, that could affect the EEG results, for example, by causing elevated alpha waves. These investigators stated that future work on the qEEG characteristics of ADHD will further aid in the accuracy of diagnosis.

Chronic Pain

Pinheiro and colleagues (2016) reviewed recent findings on EEG patterns in individuals with chronic pain. These researchers also discussed recent advances in the use of qEEG for the assessment of pathophysiology and biopsychosocial factors involved in its maintenance over time. Data collection took place from February 2014 to July 2015 in PubMed, SciELO and PEDro databases. Data from cross-sectional studies and longitudinal studies, as well as clinical trials involving chronic pain participants were incorporated into the final analysis. Primary findings related to chronic pain were an increase of theta and alpha EEG power at rest, and a decrease in the amplitude of evoked potentials after sensory stimulation and cognitive tasks. The authors concluded that increased alpha and theta power at spontaneous EEG and low amplitudes of event-related potential (ERP) during various stimuli appeared to be clinical characteristics of individuals with chronic pain; qEEG can be a simple and objective tool for studying the mechanisms involved in chronic pain, identifying specific characteristics of chronic pain conditions and providing insights about appropriate therapeutic approaches. Nevertheless, more studies are needed before drawing any conclusion on the utility of qEEG on chronic pain. Further clinical studies should be conducted to establish the clinical applicability of this instrument as an effective marker for diagnosis and guide to strategies for pain control. Systematic reviews with samples of individuals who have similar characteristics and type of pain can help determine a specific EEG pattern for each type of chronic pain.

The drawbacks of this study were:

I. data from the included studies were very heterogeneous, which prevented a meta-analysis. The conclusions were based on a qualitative analysis of the studies. Future studies should try to include similar variables, whenever possible, to allow for greater comparability of findings, and
II. the exclusion of EEG sleep studies.

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II. the exclusion of EEG sleep studies.

These researchers attempted to homogenize the sample, understanding that the awake standard EEG can be quite different from the sleep EEG. However, these findings may, in the future, be compared to findings of studies with sleeping participants in order to acquire a more comprehensive understanding of the chronic pain phenomenon. The authors noted that since they did not aim to analyze or discuss the clinical significance of EEG as a tool to detect changes after interventions, their findings and conclusions came from observational studies. Clinical trials are considered the gold standard to provide the highest level of clinical evidence. However, these researchers' questions are better addressed by the observational design. To control for quality of the evidence presented here, the articles included were assessed by criteria defined by an adapted version of the Newcastle-Ottawa scale. In general, data acquisition, processing and analysis were clearly stated in these studies, which allow reproducibility of their methods.

Prion Diseases

Franko and colleagues (2016) stated that prion diseases are universally fatal and often rapidly progressive neurodegenerative diseases; and EEG has long been used in the diagnosis of sporadic Creutzfeldt-Jakob

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disease (sCJD). However, the characteristic waveforms do not occur in all types of prion diseases. These researchers re-evaluated the utility of EEG by focusing on the development of biomarkers. They examined if abnormal qEEG parameters can be used to measure disease progression in prion diseases or predict disease onset in healthy individuals at risk of disease. In the National Prion Monitoring Cohort study, these investigators performed qEEG on 301 occasions in 29 healthy controls and 67 patients with prion disease. Patients had either inherited prion disease or sCJD. These researchers computed the main background frequency, the α and θ power and the α/θ power ratio, then averaged these within 5 electrode groups. These measurements were then compared among participant groups and correlated with functional and cognitive scores cross-sectionally and longitudinally. The authors found lower main background frequency, α power and α/θ power ratio and higher θ power in patients compared to control participants. The main background frequency, the power in the α band and the α/θ power ratio also differed in a consistent way among the patient groups. Moreover, the main background frequency and the α/θ power ratio correlated significantly with functional and cognitive scores. Longitudinally, change in these parameters also showed significant correlation with the change in clinical and cognitive scores. The authors concluded that these findings supported the use of qEEG to follow the progression of prion disease, with potential to help evaluate the treatment effects in future clinical-trials. Priorities for future work should include the use of these technologies in a clinical trial setting as an exploratory biomarker, the continued study of healthy at-risk individuals and consideration of related technologies such as magnetoencephalography.

This study had 2 major drawbacks:

I. studies of a rare disease were limited by sample size in addition to relatively small number of sCJD patients, and
II. no differences were observed between qEEG parameters in asymptomatic gene mutation carriers compared with healthy controls.

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Two interpretations were plausible

I. the EEG became abnormal several years before clinical onset, reflecting incipient neurodegeneration, but there were too few patients close to actual clinical onset in the asymptomatic inherited prion disease (aIPD) group to detect this, and
II. the EEG only became abnormal in IPD at clinical onset.

I. the EEG became abnormal several years before clinical onset, reflecting incipient neurodegeneration, but there were too few patients close to actual clinical onset in the asymptomatic inherited prion disease (aIPD) group to detect this, and
II. the EEG only became abnormal in IPD at clinical onset.

The authors stated that continued follow-up of aIPD patients and retrospective analysis of converting clinical cases may be helpful.

Also, an UpToDate review on “Diseases of the central nervous system caused by prions” (Brown and Lee, 2016) does not mention qEEG as a diagnostic tool.

Depression

Wang and colleagues (2017) examined the aberrant EEG oscillation in major depressive subjects with basal ganglia stroke with lesions in different hemispheres. Resting EEG of 16 electrodes in 58 stroke subjects, 26 of whom had post-stroke depression (13 with left-hemisphere lesion and 13 with right) and 32 of whom did not (18 with left lesion and 14 with right), was recorded to obtain spectral power analysis for several frequency bands. Multiple analysis of variance and receiver operating characteristic (ROC) curves were used

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to identify differences between post-stroke depression (PSD) and post-stroke non-depression (PSND), treating the different lesion hemispheres separately. Moreover, Pearson linear correlation analysis was conducted to test the severity of depressive symptoms and EEG indices. PSD with left-hemisphere lesion showed increased beta2 power in frontal and central areas, but PSD with right-hemisphere lesion showed increased theta and alpha power mainly in occipital and temporal regions. Additionally, for left-hemisphere lesions, beta2 power in central and right parietal regions provided high discrimination between PSD and PSND, and for right-hemisphere lesions, theta power was similarly discriminative in most regions, especially temporal regions. These researchers also explored the association between symptoms of depression and the power of abnormal bands, but found no such relationship. The authors concluded that the aberrant EEG oscillation in subjects with PSD differed between subjects with lesions of the left and right hemispheres, suggesting a complex association between depression and lesion location in stroke patients. The main drawbacks of this study were its relatively small sample size (n = 58) and the inclusion of participants with different lesions of the basal ganglia.

Screening for Depressed Mood after Stroke

Wang and colleagues (2018) examined the electrophysiological changes in post-stroke subjects with depressed mood. Resting-state electroencephalogram (rs-EEG) signals of 16 electrodes in 35 post-stroke depressed, 24 post-stroke non-depressed, and 35 age-matched healthy control subjects were analyzed by means of spectral power analysis, a qEEG measurement of different frequency bands. The relationship among depressed mood, functional status, lesion side, and post-stroke time was assessed by using variance and Spearman correlation analysis. Multiple analysis of variance was used to compare the differences among the 3 groups. Binary logistic regression analysis was used to establish a regression model to predict depressed mood in stroke subjects and to explore the association between depression and EEG band power; ROC curves were used to estimate the ability of spectral power selected by binary logistic regression to indicate depressed mood in stroke subjects. These researchers found that the hemisphere in which the lesion was located and the time since stroke onset had no effect on depressed mood. Only the patient's functional status was related to emotional symptoms; qEEG analysis revealed increased delta, theta, and beta2 power in stroke subjects with depressed mood, particularly in temporal regions. The theta and beta2 power in the right temporal area were shown to be highly sensitive to depressed mood, and these parameters showed good discriminatory ability for depressed subjects following stroke. The authors concluded that depressed mood after stroke was associated with functional status; and qEEG parameters may be a useful tool in timely screening for depressed mood after stroke.

BrainScope One System (Ahead 300) for Evaluation of Concussion / Traumatic Brain Injury

Vincent et al (2017) stated that the BrainScope Ahead 300 is designed for use by health care professionals to aid in the assessment of patients suspected of a mild traumatic brain injury (TBI). These investigators established normative data for the cognitive test component of the Ahead 300 system and examined the role of demographic factors on test performance. Healthy, community-dwelling adults between the ages of 18 and 80 recruited from 5 geographically distributed sites were administered Android versions of the ANAM Matching to Sample and Procedural Reaction Time tests that comprise the cognitive test component of the Ahead 300 system by trained personnel. Scores were correlated with age, education, and race. Age accounted for the majority of the variance in test scores with additional significant, but minor, contributions of education and race. Gender did not account for a significant proportion of the variance for either test. Based on these results, the normative data for 551 individuals were presented stratified by age. The authors concluded that these were the first available normative data for these tests when administered using the Ahead 300 system and will assist health care professionals in determining the degree to which scores on the cognitive tests reflect impaired performance.

Hanley et al (2017) noted that a brain electrical activity biomarker for identifying TBI in emergency department (ED) patients presenting with high Glasgow Coma Scale (GCS) after sustaining a head injury has shown promise for objective, rapid triage. In an observational study, these investigators prospectively evaluated the efficacy of an automated classification algorithm to determine the likelihood of being

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computed tomography (CT)-positive, in high-functioning TBI patients in the acute state. Adult patients admitted to the ED for evaluation within 72 hours of sustaining a closed head injury with GCS 12 to 15 were candidates for study. A total of 720 patients (18 to 85 years) meeting inclusion/exclusion criteria were enrolled in this validation trial at 11 U.S. EDs; GCS was 15 in 97%, with the 1st and 3rd quartiles being 15 (interquartile range [IQR] = 0) in the study population at the time of the evaluation. Standard clinical evaluations were conducted and 5 to 10 minutes of EEG was acquired from frontal and frontal-temporal scalp locations. Using an a priori derived EEG-based classification algorithm developed on an independent population and applied to this validation population prospectively, the likelihood of each subject being CT+ was determined, and performance metrics were computed relative to adjudicated CT findings. Sensitivity of the binary classifier (likely CT+ or CT-) was 92.3% (95% confidence interval [CI]: 87.8% to 95.5%) for detection of any intra-cranial injury visible on CT (CT+), with specificity of 51.6% (95% CI: 48.1% to 55.1%) and negative predictive value (NPV) of 96.0% (95% CI: 93.2% to 97.9%). Using ternary classification (likely CT+, equivocal, likely CT-) demonstrated enhanced sensitivity to traumatic hematomas (greater than or equal to 1 ml of blood), 98.6% (95% CI: 92.6% to 100.0%), and NPV of 98.2% (95% CI: 95.5% to 99.5%). The authors concluded that using an EEG-based biomarker high accuracy of predicting the likelihood of being CT+ was obtained, with high NPV and sensitivity to any traumatic bleeding and to hematomas; specificity was significantly higher than standard CT decision rules. They stated that the short time to acquire results and the ease of use in the ED environment suggested that EEG-based classifier algorithms have potential to impact triage and clinical management of head-injured patients.

Hack et al (2017) compared the predictive power using that algorithm (which includes loss of consciousness [LOC] and amnesia) to the predictive power of LOC alone or LOC plus traumatic amnesia. ED patients 18 to 85 years presenting within 72 hours of closed head injury, with GSC 12 to 15, were study candidates. A total of 680 patients with known absence or presence of LOC were enrolled (145 CT+ and 535 CT- patients); 5 to 10 mins of eyes closed EEG was acquired using the Ahead 300 hand-held device, from frontal and fronto-temporal regions. The same classification algorithm methodology was used for both the EEG-based and the LOC-based algorithms. Predictive power was evaluated using area under the ROC curve (AUC) and odds ratios (ORs). The quantitative EEG (QEEG)-based classification algorithm demonstrated significant improvement in predictive power compared with LOC alone, both in improved AUC (83% improvement) and OR (increase from 4.65 to 16.22). Adding RGA and/or PTA to LOC was not improved over LOC alone. The authors concluded that rapid triage of TBI relies on strong initial predictors. Addition of an EEG-based marker was shown to out-perform report of LOC alone or LOC plus amnesia, in determining risk of an intra-cranial bleed. Moreover, they stated that ease of use at point-of-care, non-invasive, and rapid result using such technology suggested significant value added to standard clinical prediction.

Hanley et al (2018) stated that the potential clinical utility of a novel QEEG-based Brain Function Index (BFI) as a measure of the presence and severity of functional brain injury was studied as part of an independent prospective validation trial. The BFI was derived using QEEG features associated with functional brain impairment reflecting current consensus on the physiology of concussive injury. A total of 720 adult patients (18 to 85 years of age) evaluated within 72 hours of sustaining a closed head injury were enrolled at 11 U.S. EDs; GCS score was 15 in 97%. Standard clinical evaluations were conducted and 5 to 10 mins of EEG acquired from frontal locations. Clinical utility of the BFI was assessed for raw scores and percentile values. A multi-nomial logistic regression analysis demonstrated that the ORs (computed against controls) of the mild and moderate functionally impaired groups were significantly different from the OR of the CT-positive (CT+, structural injury visible on CT) group ($p = 0.0009$ and $p = 0.0026$, respectively). However, no significant differences were observed between the ORs of the mild and moderately functionally impaired groups. Analysis of variance (ANOVA) demonstrated significant differences in BFI among normal (16.8%), mild TBI (mTBI)/concussed with mild or moderate functional impairment, (61.3%), and CT+ (21.9%) patients (p potential to aid in early diagnosis and thereby potential to impact the sequelae of TBI by providing an objective marker that is available at the point-of-care, hand-held, non-invasive, and rapid to obtain.

Conley and co-workers (2019) noted that sports-related concussion is associated with a range of short-term functional deficits that are commonly thought to recover within a 2-week post-injury period for most, but certainly not all, persons; and rs-EEG may prove to be an affordable, accessible, and sensitive method of assessing severity of brain injury and rate of recovery after a concussion. These investigators presented a

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systematic review of rs-EEG in sports-related concussion. A systematic review of articles published in the English language, up to June 2017, was retrieved via PsychINFO, Medline, Medline In Process, Embase, SportDiscus, CINAHL, and Cochrane Library, Reviews, and Trials. The following key words were used for database searches: electroencephalography, quantitative electroencephalography, qEEG, cranio-cerebral trauma, mild traumatic brain injury, mTBI, traumatic brain injury, brain concussion, concussion, brain damage, sport, athletic, and athlete. Observational, cohort, correlational, cross-sectional, and longitudinal studies were all included in the current review. A total of 16 articles met inclusion criteria, which included data on 504 athletes and 367 controls. All 16 articles reported some abnormality in rs-EEG activity after a concussion; however, the cortical rhythms that were affected varied. The authors concluded that despite substantial methodological and analytical differences across the 16 studies, the current review suggested that rs-EEG may provide a reliable technique to identify persistent functional changes in athletes after a concussion. Moreover, they stated that because of the varied approaches, however, considerable work is needed to establish a systematic methodology to assess its efficacy as a marker of return-to-play.

Furthermore, an UpToDate review on “Acute mild traumatic brain injury (concussion) in adults” (Evans and Whitlow, 2018) does not mention BrainScope / EEG-based technology as a diagnostic tool.

Tenney and colleagues (2021) stated that despite many decades of research, controversy regarding the use of qEEG for the accurate diagnosis of mild TBI (mTBI) remains. These investigators noted that this guideline on “Use of quantitative EEG for the diagnosis of mild traumatic brain injury’ from the Committee of the American Clinical Neurophysiology Society is meant to aid clinicians by providing an expert review of the clinical usefulness of qEEG techniques for the diagnosis of mTBI. It addressed the following primary aim: For patients with or without post-traumatic symptoms (abnormal cognition or behavior), does qEEG either at the time of injury or remote from the injury, as compared with current clinical diagnostic criteria, accurately identify those patients with mTBI (i.e., concussion)? Secondary aims included differentiating between mTBI and other diagnoses, detecting mTBI in the presence of central nervous system (CNS) medications, and pertinence of statistical methods for measurements of qEEG components. The authors found that for patients with or without symptoms of abnormal cognition or behavior, current evidence does not support the clinical use of qEEG either at the time of the injury or remote from the injury to diagnose mTBI (level U). Furthermore, the evidence does not support the use of qEEG to differentiate mTBI from other diagnoses or detect mTBI in the presence of CNS medications, and suitable statistical methods do not exist when using qEEG to identify patients with mTBI. Based upon the current literature review, these researchers stated that qEEG remains an investigational tool for mTBI diagnosis (Class III Evidence).

Psychiatric Disorders

McVoy and colleagues (2019) noted that qEEG has emerged as a potential intermediate biomarker for diagnostic clarification in mental illness. In a systematic review, these researchers examined published studies that used qEEG in youth with psychiatric illness between 1996 and 2017. They conducted a comprehensive database search of CINAHL, PubMed, and Cochrane using the following keywords: "quantitative EEG" and depression (MDD), anxiety, attention-deficit hyperactivity disorder (ADHD), autism spectrum disorder (ASD), eating disorder, conduct, substance use, schizophrenia, post-traumatic stress disorder, and panic disorder. The search yielded 516 titles; 33 met final inclusion criteria, producing a total of 2,268 youth aged 4 to 18 years. qEEG was most frequently studied as a potential diagnostic tool in pediatric mental illness; few studies assessed therapeutic response. Studies showed higher theta/beta ratio in ADHD versus healthy controls (HC). The most consistent finding in ASD was decreased coherence in ASD versus HC. Studies showed MDD has lower temporal coherence and inter-hemispheric coherence in sleep EEGs than HC. The authors concluded that further research is needed in the areas of mood, anxiety, ASD, and relationship to treatment. It remained unknown if abnormalities in qEEG were non-specific markers of pediatric psychiatric illness or if they have the potential to differentiate types of psychopathology.

Perera and associates (2019) stated that obsessive-compulsive disorder (OCD) is a chronic disease that causes significant decline in the quality of life (QOL) of afflicted individuals. Due to the limited understanding of the underlying pathophysiology of OCD, successful treatment remains elusive. Although many have studied the pathophysiology of OCD through EEG, limited attempts have been made to synthesize and interpret their findings. To bridge this gap, these researchers conducted a comprehensive

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literature review using Medline/PubMed and considered the 65 most relevant studies published before June 2018. The findings were categorized into qEEG, sleep-related EEG and ERPs. Increased frontal asymmetry, frontal slowing and an enhancement in the ERP known as error-related negativity (ERN) were consistent findings in OCD. However, sleep EEG and other ERP (P3 and N2) findings were inconsistent. Additionally, these investigators analyzed the usefulness of ERN as a potential candidate endophenotype. They hypothesized that dysfunctional frontal circuitry and over-active performance monitoring were the major underlying impairments in OCD. Furthermore, these researchers conceptualized that defective fronto-striato-thalamic circuitry causing poor cerebral functional connectivity gave rise to the OCD behavioral manifestations. Finally, the authors discussed transcranial magnetic stimulation and EEG (TMS-EEG) applications in future research to further the understanding of the underlying pathophysiology of OCD.

Furthermore, UpToDate reviews on "Eating disorders: Overview of epidemiology, clinical features, and diagnosis" (Yager, 2019), “Obsessive-compulsive disorder in adults: Epidemiology, pathogenesis, clinical manifestations, course, and diagnosis” (Simpson, 2019) and "Posttraumatic stress disorder in adults: Epidemiology, pathophysiology, clinical manifestations, course, assessment, and diagnosis" (Sareen, 2019) do not mention qEEG as a management tool.

Anxiety

Imperatori and associates (2019) stated that although several researches examined Default Mode Network (DMN) alterations in individuals with anxiety disorders, up to now no studies have examined DMN functional connectivity in non-clinical individuals with high-trait-anxiety using qEEG. These investigators extended previous findings examining the association between trait anxiety and DMN EEG functional connectivity. A total of 23 individuals with high-trait-anxiety and 24 controls were enrolled; EEG was recorded during 5 mins of resting state (RS). EEG analyses were conducted by means of the exact Low-Resolution Electromagnetic Tomography software (eLORETA). Compared to controls, individuals with high-trait-anxiety showed a decrease of theta connectivity between right medial prefrontal cortex (mPFC) and right posterior cingulate/retrosplenial cortex. A decrease of beta connectivity was also observed between right mPFC and right anterior cingulate cortex. Furthermore, DMN functional connectivity strength was negatively related with STAI-T total score (i.e., lower connectivity was associated with higher trait anxiety), even when controlling for potential confounding variables (i.e., sex, age, and general psychopathology). The authors concluded that the findings of this study suggested that high-trait-anxiety individuals failed to synchronize DMN during RS, reflecting a possible top-down cognitive control deficit. These results may help in the understanding of the individual differences in functional brain networks associated with trait anxiety, a crucial aim in the prevention and in the early etiology understanding of clinical anxiety and related sequelae. Moreover, these researchers stated that small sample size (n = 23 for subjects with anxiety) made it difficult to draw definitive conclusions. Furthermore, they did not evaluate state variation of anxiety, which made their interpretation specific to trait anxiety.

Bong and colleagues (2020) noted that the Research Domain Criteria (RDoC) project was proposed by the National Institute of Mental Health in 2010 to create a new diagnostic system including symptoms and data from genetics, neuroscience, physiology, and self-reports. These researchers determine the link between anxiety and executive functions via qEEG based on the RDoC system; 19-channel EEGs were recorded at the psychiatric clinic from 41 patients with symptoms of anxiety. The EEG power spectra were analyzed. The Executive Intelligence Test (EXIT) including the K-WAIS-IV, Stroop, controlled oral word association, and the design fluency tests were carried out. A partial, inversed, and significant association was observed between executive intelligence quotient (EIQ) and the absolute delta power in the central region. Similarly, a partial, inversed, and significant association was seen between design fluency and the absolute delta power in the left parietal area. The authors concluded that the findings of this study suggested that the increase in delta power in the central region and left P3 was negatively correlated with the decrease in executive function. It is expected that the absolute delta power plays a specific role in the task-negative default mode network in the relationship between anxiety and executive function.

The authors stated that this study had several drawbacks. First, since it was conducted without a control group, it was difficult to specify this as a result that occurred only in individuals who complained of anxiety; thus, in future studies, there is a need to examine if an increase in absolute delta power in anxiety and

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executive function appears only when complaining of anxiety using a healthy control. Second, this study's sample size was relatively small (n = 41), of which only 11 were male. A small number of subjects may be correlated with this study's results showing negative findings regarding other indicators except for delta waves; thus, future research should include more subjects to obtain more accurate and specific results. Third, DMN appeared to play a role in the relationship between anxiety and executive function; however, additional studies are needed to confirm this. Fourth, these researchers did not find a relationship between anxiety and EEG findings, nor between anxiety and cognitive function. This may be due to the high, clinical level of anxiety among the subjects. In this study, only participants who had 40 points or more on either subscale of the Korean version of State-Trait Anxiety Inventory (STAI) were included, and no control group was used. The means of both the STAI (state) and STAI (trait) were very high at 59.63 ± 11.74 and 59.54 ± 10.86, respectively, which may have introduced a bias. Therefore, the methods limited the ability to test the hypothesis that as anxiety levels increase there will be changes in qEEG. Given the limitation of the method, it may be premature to conclude that there is no correlation between severity of anxiety and EEG analyses. These researchers stated that further studies, with controls, are needed to evaluate the correlation between anxiety severity and qEEG. Fifth, this study was based on a single local university medical center. Patients who visited the hospital often complained of severe anxiety at a pathological level. As a result of this study, it is difficult to explain the normal range of anxiety.

Furthermore, an UpToDate review on "Generalized anxiety disorder in adults: Epidemiology, pathogenesis, clinical manifestations, course, assessment, and diagnosis" (Baldwin, 2020) does not mention qEEG or brain mapping as a management tool.

Prediction of Outcomes in Children Following Cardiac Arrest

In a prospective, single-center, observational study, Lee and colleagues (2019) qEEG features predict neurologic outcomes in children following cardiac arrest. This trial included 87 consecutive children who were resuscitated and admitted to the pediatric ICU following cardiac arrest. Full-array conventional EEG data were obtained as part of clinical management. These researchers computed 8 qEEG features from 5-min epochs every hour after return of circulation. They developed predictive models utilizing random forest classifiers trained on patient age and 8 qEEG features to predict outcome. The features included SD (used as a feature to illustrate the variance of the signal) of each EEG channel, normalized band power in alpha, beta, theta, delta, and gamma wave frequencies, line length, and regularity function scores. These investigators measured outcomes using Pediatric Cerebral Performance Category (PCPC) scores. They evaluated the models using 5-fold cross-validation and 1,000 bootstrap samples. The best performing model had a 5-fold cross-validation accuracy of 0.8 (0.88 area under the receiver operating characteristic curve). It had a positive predictive value of 0.79 and a sensitivity of 0.84 in predicting patients with favorable outcomes (PCPC score of 1 to 3). It had a negative predictive value of 0.8 and a specificity of 0.75 in predicting patients with unfavorable outcomes (PCPC score of 4 to 6). The model also identified the relative importance of each feature. Analyses using only frontal electrodes did not differ in prediction performance compared to analyses using all electrodes. The authors concluded that qEEG features can standardize EEG interpretation and predict neurologic outcomes in children following cardiac arrest. Moreover, these researchers stated that further study is needed to examine the benefit of using qEEG features in the context of multi-modal models for clinical trials or neuro-prognostication.

The authors stated that while the findings of this study were promising, there were several drawbacks. First, this study had a relatively small sample size (n = 69); thus, these investigators were unable to partition their data to include a holdout validation set for their model. While these researchers used cross-validation and bootstrapping to overcome the limitations of the small sample size, these findings should be validated in a larger study. Second, this study used a MATLAB script for artifact removal that was not as accurate as clinically trained neurologists. While manual EEG review is the current gold standard, independent component analysis–based algorithms could obtain EEG of improved signal quality. Third, this study was based only on EEG, but multi-modal models also incorporating clinical features could provide a more complete understanding of the neurologic state of the patient and yield more accurate neuro-prognostication. Fourth, these investigators only examined short-term outcome using a relatively simple outcome measure; future studies might incorporate longer-term patient-centered neurobehavioral outcomes. Fifth, clinically

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interpreted EEG data were known to the clinical teams and may have influenced care decisions. To reduce the influence of individual decisions regarding withdrawal of technological support on outcome categorization, these researchers used an outcome measure that grouped death and unfavorable neurologic outcome. Therefore, outcome would be categorized as unfavorable whether a family chose to withdraw or continue technological support of a child with severe neurologic injury.

Quantitative EEG as a Prognostic Biomarker for Long-Term Outcomes in Preterm Infants with Neurological and Medical Complications

Cainelli and associates (2021) noted that prematurity is a prototype of biological risk that could affect the late neurocognitive outcome; however, the condition itself remains a non-specific marker. Precise prognostic instruments are lacking, mostly in patients with low-grade conditions. In a longitudinal, 6-year study, these researchers examined the prognostic role of neonatal spectral EEG in premature infants without neurological complications. The study cohort was 26 children born 23 to 34 weeks gestational ages; all neonates underwent multi-channel EEG recordings at 35 weeks post-conception. EEG data were transformed into the frequency domain and divided into delta (0.5 to 4 Hz), theta (5 to 7 Hz), alpha (8 to 13 Hz), and beta (14 to 20 Hz) frequency bands. At 6 years, a neuropsychological and behavioral evaluation was carried out. Correlations between spectral bands and neuropsychological assessments were carried out with a conservative and robust Bayesian correlation model using weakly informative priors. The correlation of neuropsychological tasks to spectral frequency bands highlighted a significant association with visual and auditory attention tests. The performance on the same tests appears to be mainly impaired. The authors found that spectral EEG frequencies were independent predictors of performance in attention tasks. These researchers hypothesized that spectral EEG might reflect early circuitries' imbalance in the reticular ascending system and cumulative effect on ongoing development, pointing to the importance of early prognostic instruments. They stated that longitudinal long-term studies are scarce but crucial for the inferential attributive process. These investigators stated that future research should examine the use of spectral EEG in premature neonates with neurological and medical complications.

The authors stated that this study had several drawbacks. First, the sample size was very small (n = 26)l; this long-term longitudinal study requested the need for covering a long period (from the neonatal period to the age when complex cognitive functions developed and could be tested). In 6 years, some patients dropped out or became untraceable. However, given the high effort in the recruitment procedure, the percentage of parents who refused to participate at this stage of the follow-up was very low; thus, the majority of lost subjects were due to logistic reasons and not the choice of parents, often biased by the effective outcome of their child. Second, the statistical approach was reliable even with small sample sizes, but, without other confirmatory studies, the small number of patients could limit the generalizability of these findings. Third, the high number of potential confounding variables may have influenced the outcome in the 6 years of the life of the child. For example, these researchers could have examined parental mental state, known to potentially bias assessment of children’s health.

Finally, we selected a group of patients with no evident neurological risk factors other than the prematurity itself. Therefore, our results cannot be generalized to the entire population of preterm infants. However, we were specifically interested in these children in whom the outcome is highly uncertain. In premature infants with evident signs of neurological dysfunctions, the prognosis is relatively simpler. Visual inspections of EEG, MRI, and clinical evaluation may help clinicians in the diagnosis.

In a systematic review, van 't Westende and colleagues (2021) examined qEEG measures as predictors of long-term neurodevelopmental outcome in infants with a post-conceptional age below 46 weeks, including typically developing infants born at term, infants with heterogeneous underlying pathologies, and infants born pre-term. These investigators carried out a comprehensive search using PubMed, Embase, and Web of Science from study inception up to January 8, 2021. Studies that examined associations between neonatal qEEG measures, based on conventional and amplitude-integrated EEG, and standardized neurodevelopmental outcomes at 2 years of age or older were reviewed. Significant associations between neonatal qEEG and long-term outcome measures were grouped into 1 or more of the following categories: cognitive outcome; motor outcome; composite scores; and other standardized outcome assessments. A total

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of 24 out of 1,740 studies were included. Multiple studies showed that conventional EEG-based absolute power in the delta, theta, alpha, and beta frequency bands and conventional and amplitude-integrated EEG-related amplitudes were positively associated with favorable long-term outcome across several domains, including cognition and motor performance. Furthermore, a lower presence of discontinuous background pattern was also associated with favorable outcomes. However, interpretation of the results was limited by heterogeneity in study design and populations. The authors concluded that neonatal quantitative EEG measures may be used as prognostic biomarkers to identify those infants who will develop long-term difficulties and who might benefit from early interventions. Moreover, these researchers stated that these findings underscored that qEEG analysis has the potential to improve neonatal care since it can provide new prognostic information in infants at risk; however, inference is limited by heterogeneity in populations and study designs. They stated that there is a definite need for more explorative studies to identify the most relevant qEEG measures for prognostication in clinical practice.

The authors stated that a main drawback of this review was the lack of a meta-analysis, which was impeded by heterogeneity in study design. The studies included in this review examined different study populations (e.g., infants born preterm, infants with asphyxia, or infants with tuberous sclerosis complex). Furthermore, EEG settings and analyses were substantially different across studies, including differences in the number of electrodes, reference electrodes, and qEEG measures used. This precluded determination of the best qEEG measures as prognostic biomarkers, indicating the need to standardize neonatal EEG acquisition and analysis. Moreover, a wide range of long-term outcome assessments were used, even within single outcome categories. Assessment designs were diverse, including neurological examinations, parental questionnaires, and standardized batteries. For example, cognition was evaluated by 9 different outcome assessments (Bielefelder screening, Wechsler Preschool and Primary Scales of Intelligence, Wechsler Intelligence Scale for Children, NEuroPSYchological Assessment, BSID-II and Bayley-III, Kaufman Assessment Battery for Children, Rey Auditory Verbal Learning Test, Behavior Rating Inventory of Executive Function, and Stanford–Binet Test) across 16 studies. In addition, age at follow-up varied widely and there was a lack of standard definitions as to what should be regarded as an adverse outcome. Since these investigators examined associations, the definition of adverse outcome might not have influenced the interpretation of the observed trends. Nonetheless, heterogeneity in definitions impeded the insight into clinical relevancy of the results of the current review, which should be considered during interpretation. Another drawback was the lack of large high-quality studies. Furthermore, a risk of bias should be taken into account when interpreting the results because at least 21 of 24 studies had no consecutive and/or complete inclusion of patients. Moreover, different guidelines exist for conducting systematic reviews, each with their own strengths and weaknesses. For example, when compared to the AMSTAR checklist, a limitation of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline is that a single reviewer is allowed to perform data extraction, which might increase the risk for subjective influences in the extracted data.

Quantitative EEG in COVID-19 Patients

Pati and associates (2020) noted that there is an unmet need for biomarkers in monitoring therapy responses and prognosticating neurological recovery in comatose patients with coronavirus disease 2019 (COVID-19). Acute encephalopathy is increasingly recognized in critically ill mechanically ventilated patients with COVID-19. As brain injury is often the principal determinant of functional outcomes in critically ill patients, these researchers examined if cortical electrophysiological markers could prognosticate neurological recovery. Continuous EEG (cEEG) allows non-invasive monitoring of neural activity over time that can permit prognostication in real-time at the bedside. In a retrospective, single-center study, these investigators tested the hypothesis that qEEG features extracted from cEEG could predict neurological outcome in critically ill patients with COVID-19 after sedation is withdrawn.

A total of 10 consecutive patients (mean age of 61.3 years) met the inclusion criteria: mechanically ventilated and critically ill; polymerase chain reaction confirmed COVID-19; had cEEG monitoring (21 channels sampled at 250 Hz) over 48 hours; and had a definitive outcome at discharge as determined using Cerebral Performance Category Scale (CPC). Outcomes were grouped into good (CPC less than 2) and poor (CPC 3 to 5). Multiple epochs of EEG collected over 37 patient-days were labeled for analysis as follows: EEG reactivity -- 40 seconds epoch following voice and noxious-sensory stimuli; and baseline -- epochs prior to

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reactivity. As part of the approved institutional protocol, EEG reactivity was tested after sedative and paralytic medications were withheld for clinical examination. The sound stimuli included calling out the patients' names and clapping. Tactile and noxious stimuli included sternal rub, trapezius pressure, and nose stimulation with a swab. The epileptologists reported EEG reactivity as “present” or “absent” or “indeterminate". If a change in the EEG frequency or amplitude was present post stimuli, the epileptologists reported as positive EEG reactivity. The authors concluded that qEEG features at baseline and reactivity could prognosticate neurological recovery in critically ill patients with COVID-19. The ubiquity of cEEG monitoring allows rapid translation in the clinical practice to facilitate decision-making to mobilize or withhold limited resources, guide patient selection, and planning adaptive clinical trials. Moreover, these researchers stated that more data in larger prospectively studied cohorts are needed to corroborate these findings.

Kopanska and colleagues (2021) stated that the SARS-CoV-2 virus is able to cause abnormalities in the functioning of the nervous system and induce neurological symptoms with the features of encephalopathy, disturbances of consciousness and concentration and a reduced ability to sense taste and smell as well as headaches. One of the methods of detecting these types of changes in COVID-19 patients is an EEG test, which allows information to be obtained regarding the functioning of the brain as well as diagnosing diseases and predicting their consequences. These investigators reviewed the latest research on changes in EEG in patients with COVID-19 as a basis for further qEEG diagnostics and EEG neurofeedback training. Based on the available scientific literature using the PubMed database from 2020 and early 2021 regarding changes in the EEG records in patients with COVID-19, a total of 17 publications were included in the analysis. In patients who underwent an EEG test, changes in the frontal area were observed. A few patients were not found to be responsive to external stimuli. Furthermore, a previously non-emerging, uncommon pattern in the form of continuous, slightly asymmetric, monomorphic, biphasic and slow delta waves occurred. The authors concluded that the results of this analysis indicated that the SARS-CoV-2 virus causes changes in the nervous system that can be manifested and detected in the EEG record. The small number of available articles, the small number of research groups and the lack of control groups suggested the need for further research regarding the short- and long-term neurological effects of the SARS-CoV-2 virus and the need for unquestionable confirmation that observed changes were caused by the virus per se and did not occur before. The presented studies described non-specific patterns appearing in encephalograms in patients with COVID-19. Moreover, these researchers stated that one important obstacle apart from the number of patients available for analysis was the heterogeneity of the EEG techniques used and certain subjectivity of their interpretation; therefore, the consequent implementation of qEEG techniques would enable a more precise, objective description of these changes and would create a solid base for attempts to implement qEEG neurofeedback therapy as a part of comprehensive treatment for patients who aspire to improve their neurological symptoms acquired in connection with the SARS-CoV-2 virus infection.

Furthermore, an UpToDate review on “COVID-19: Evaluation and management of adults following acute viral illness” (Mikkelsen and Abramoff, 2021) does not mention quantitative EEG (qEEG) as a management option.

Quantitative EEG as a Biomarker in Non-Invasive Brain Stimulation Therapy in Patients with Parkinson's Disease

de Carvalho Costa e al (2022) noted that PD is a progressive neurodegenerative disorder characterized by motor and non-motor symptoms, aside from alterations in the EEG already registered. Non-invasive brain stimulation (NIBS) techniques have been suggested as an alternative rehabilitative therapy; however, the neurophysiological changes associated with these techniques are still unclear. In a systematic review, these investigators identified the nature and extent of research evidence on the effects of NIBS techniques in the cortical activity measured by EEG in patients with PD. A systematic scoping review was configured by gathering evidence on the following bases: PubMed (Medline), PsycINFO, ScienceDirect, Web of Science, and cumulative index to nursing & allied health (CINAHL). They included clinical trials with patients with PD treated with NIBS and examined by EEG pre-intervention and post-intervention. These researchers employed the criteria of Downs and Black to evaluate the quality of the studies. Repetitive transcranial magnetic stimulation (TMS), transcranial electrical stimulation (tES), electrical vestibular stimulation, and

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binaural beats (BBs) are non-invasive stimulation techniques used for the treatment of cognitive and motor impairment in patients with PD. This systematic scoping review found that the current evidence suggested that NIBS could change quantitative EEG in patients with PD. However, considering that the quality of the studies varied from poor-to-excellent, the low number of studies, variability in NIBS intervention, and quantitative EEG measures, these investigators were not yet able to use the EEG outcomes to predict the cognitive and motor treatment response following brain stimulation. The authors concluded that based on these findings, they recommended that additional studies are needed to validate EEG as a biomarker in non-invasive brain stimulation therapy in patients with PD.

The authors stated that the main drawback of this systematic review was the heterogeneity of protocols between the included studies could somehow limit the conclusion. Moreover, a high risk of bias was present in several studies, which called for caution in interpreting these findings. There were multiple sources of potential heterogeneity within the EEG and brain stimulation literature relating to the variability in stimulation parameters and outcomes measured, dose, and clinical characteristics. One of the main factors lacking in 50 % of the studies was robust concordance regarding the enhancement of motor recovery associated with the clinical application of brain stimulation and EEG. Moreover, completeness of evidence was lacking regarding electrophysiological markers reflecting tDCS effects and cognitive outcomes in PD. This is an important factor to take into account when talking about brain modulation techniques and progressive impairment. This diversity of metrics and the lack of clear underlying hypotheses regarding the electrophysiology of motor and cognitive parameters made it hard to interpret the effect of treatment. There is currently insufficient high-quality evidence to draw conclusions regarding the benefits or harms of NIBS and electrophysiologic correlates on PD.

Diagnosis of Pathological Fatigue

Heitmann et al (2023) noted that fatigue is a highly prevalent and disabling symptom of many disorders and syndromes, resulting from different pathomechanisms. However, whether and how different mechanisms converge and result in similar symptomatology is only partially understood, and trans-diagnostic biomarkers that could further the diagnosis and treatment of fatigue are lacking. These investigators carried out a trans-diagnostic systematic review of quantitative resting-state EEG and magnetoencephalography (MEG) studies in adult patients suffering from pathological fatigue in different disorders. Studies examining fatigue in healthy subjects were excluded. The risk of bias was assessed using a modified Newcastle-Ottawa Scale. Semi-quantitative data synthesis was carried out using modified albatross plots. After searching Medline, Web of Science Core Collection, and Embase, a total of 26 studies were included. Cross-sectional studies showed increased brain activity at theta frequencies and decreased activity at alpha frequencies as potential diagnostic biomarkers. However, the risk of bias was high in many studies and domains. The authors concluded that this trans-diagnostic systematic review synthesized evidence on how resting-state M/EEG might serve as a diagnostic biomarker of pathological fatigue. These researchers stated that this review might aid in guiding future magneto-/EEG studies on the development of fatigue biomarkers.

Prediction of Clinical Impairment in Stroke

Lanzone et al (2023) stated that qEEG has shown promising results as a predictor of clinical impairment in stroke. In a systematic review, these investigators examined studies that focused on qEEG metrics in the resting EEG of patients with mono-hemispheric stroke, summarized current knowledge and paved the way for future research. Following the PRISMA guidelines, these researchers systematically searched the literature for studies that fitted the inclusion criteria. Rayyan QCRR was used to allow de-duplication and collaborative blinded paper review. Due to multiple outcomes and non-homogeneous literature, a scoping review approach was used to address the topic. The initial search (PubMed, Embase, Google scholar) yielded 3,200 papers. After proper screening, these investigators selected 71 papers that fitted the inclusion criteria and they developed a scoping review thar described the current state of the art of qEEG in stroke. Notably, among selected papers, 53 (74.3 %) focused on spectral power; 11 (15.7 %) focused on symmetry indexes, 17 (24.3 %) on connectivity metrics, while 5 (7.1 %) were about other metrics (e.g., detrended fluctuation analysis). Moreover, 42 (58.6 %) studies were carried out with standard 19 electrodes EEG caps and only a minority used high-definition EEG. The authors concluded that they systematically evaluated

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major findings on qEEG and stroke, evidencing strengths, and potential pitfalls of this promising branch of research.

Furthermore, UpToDate reviews on “Initial assessment and management of acute stroke” (Oliveira-Filho and Mullen, 2024), and “Overview of the evaluation of stroke” (Caplan, 2024) do not mention qEEG as a management option.

Marker of Central Neuropathic Pain Intensity in Spinal Cord Injury

Nawaz et al (2024) stated that it is well known from cross-sectional studies that pain intensity affects brain activity as measured by EEG in patients with neuropathic pain (NP); however, quantitative characterization of EEG is scarce. In a longitudinal, pilot study, 10 patients with spinal cord injury (SCI)-related NP recorded their home EEG activity 10 days before and after taking medications over a period of several weeks. The reduction in pain due to medications was accompanied by changes in the resting state EEG and its reactivity to eyes opening (EO) and closing (EC). There was a significant positive correlation between the frontal theta band and the intensity of pain (visual numerical scale [VNS]) pre-medication ($p = 0.007$, Pearson $R = 0.29$) and theta, alpha, and lower beta (6 to 15 Hz) band power and the intensity of pain following post-medication over the frontal, central, and parietal cortices. Reactivity had a negative correlation with pain intensity at all locations and frequency bands and demonstrated similar behavior in wider frequency bands like 8 to 15 Hz at the occipital cortex and 2 to 12 Hz at the frontal cortex. The authors concluded that EEG could be used to detect the intensity of NP to serve as a surrogate or pharmacodynamic marker. Moreover, these researchers stated that the feasibility of this approach even with highly disabled users opens the door to larger-scale studies of different neurological conditions. Well-designed studies are needed to validate these preliminary findings of a pilot study. In addition, the authors stated that future longitudinal studies with a larger number of subjects should be able to reveal additional smaller effect sizes relationships between pain and EEG, which could not be detected in the study.

The authors stated that an obvious drawback of this and other similar studies was that a VNS was employed to report the intensity of pain, although it is highly subjective; thus, might vary greatly among individuals. In this pilot study, this issue was somewhat counterbalanced by repeated measures from the same subjects. Furthermore, in order to minimize inter-subject variability, these investigators also computed the relative change in EEG power and pain post-medication with respect to pre-medication. No significant correlation was observed. The reason for this might be that, with only 10 subjects, these researchers did not have sufficient power to notice the effect, which even without correction for multiple comparison revealed significant correlation in the central lower beta band only. Studies of different types of chronic pain showed that pain relief due to medications or interventions is accompanied by changes in EEG, although the direction of change was inconclusive and the regression analysis of longitudinal studies were missing.

Management of Patients After Chimeric Antigen Receptor (CAR)-T Cell Therapy

Matthews et al (2025) stated that neurotoxicity, encephalopathy, and seizures could occur after chimeric antigen receptor (CAR)-T cell therapy. These researchers examined the value EEG offers for patients undergoing CAR-T treatment in clinical practice, including possible diagnostic, management, and prognostic roles. All patients developing CAR-T related neurotoxicity referred for EEG were eligible for inclusion. Reasons for EEG referral and qualitative EEG findings were analyzed and reported. The relationship between objective qEEG encephalopathy grade and clinical neurotoxicity (immune effector cell-associated neurotoxicity syndrome; ICANS) grade was determined. The prognostic ability of qEEG grade was evaluated for survival and functional status. A total of 28 patients with 53 EEG recordings were included. Common reasons given on EEG referrals were possible seizure diagnosis ($n = 38$), reduced consciousness ($n = 8$), and superimposed cerebral infection ($n = 4$). A total of 4 focal seizures were detected on 3 (3/53; 5.7 %) EEGs. There was a moderately positive correlation between qEEG grade and ICANS grade ($r = + 0.41$, $p = 0.030$). qEEG grade could not predict survival at 3 months (AUC = 0.673) or 6 months (AUC = 0.578), nor could it predict functional status at 1 month ($r = + 0.40$; $p = 0.080$), 3 months ($r = + 0.19$; $p = 0.439$), or time to return to baseline ($r = + 0.32$; $p = 0.156$). The authors concluded that EEG was useful in seizure diagnosis; and qEEG has a possible role as a specific biomarker of encephalopathy/neurotoxicity. Moreover,

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these researchers stated that EEG generated no tangible changes in patient management; and qEEG was unable to prognosticate survival or functional status.

Management of Patients in Coma

Degano et al (2025) stated that recent studies on qEEG in coma has proposed several metrics correlating with consciousness level; however, the heterogeneous nature of coma could challenge the generalizability of these measures. In a retrospective study, these investigators examined of alpha-coma, an electro-clinical pattern characterized by a widespread, non-reactive alpha rhythm often linked to poor outcomes. The objective of this trial was to quantify the electrophysiological features of alpha-coma and compared them to the alpha rhythm in awake HC, seeking clearer insights into qEEG analysis in comatose states. A total of 14 alpha-coma patients were selected from University Hospitals of Geneva and age-matched with 14 HC from an open-source data-set. EEG data were pre-processed and analyzed to extract power spectra, spectral decay (aperiodic activity), sample entropy, and functional connectivity. Alpha-coma patients did not differ in alpha power but exhibited significantly higher levels of spectral decay ($p < 0.001$), suggesting a convergence toward an inhibitory state. Sample entropy was significantly higher in alpha-coma patients ($p = 0.01$), indicating an increase in the cortical complexity in alpha-coma compared with HC. The authors concluded that alpha-coma demonstrated increased aperiodic activity and EEG complexity, despite similar alpha power and clustering coefficient. The increased aperiodic activity aligned with findings in other comatose patients, including those sedated or with subcortical dysfunction. However, the increased entropy contradicted existing literature, suggesting that alpha-coma may represent a state of widespread cortical dysfunction likely resulting from non-hierarchical, turbulent brain activity. This observation indicated that the loss of consciousness does not guarantee consistent cortical measures across the whole spectrum of EEG patterns. Moreover, these researchers proposed that the use of entropy as a measure of overall cognitive-related brain complexity should be approached with caution, underscoring the importance of considering multiple measures when evaluating level of consciousness.

The authors stated that this study had several drawbacks. First, because of the rarity of the alpha-coma condition, the total sample size was relatively small ($n = 14$), and there was a lack of homogeneity in etiology. In addition, because of poor clinical outcomes among patients with alpha-coma, there was insufficient variability in the sample to examine if measures of brain complexity or aperiodic activity could serve as reliable predictors of clinical outcomes. Second, the focus on the alpha frequency range also complicated direct comparisons with other comatose populations, where different EEG frequency ranges (e.g., alpha-theta, delta) might be more relevant. Nevertheless, based on these findings and what has been reported in the literature for selected populations of patients, these investigators proposed a conceptual scheme that generates a new interpretation of these quantitative cortical measures. Third, wakeful HCs exhibited a balanced neuronal activity, whereas sedated and comatose patients displayed a predominant inhibitory pattern highlighted by an increase in the slope exponent of the power spectrum. In entropy, wakeful subjects showed normal brain complexity, whereas comatose patients with primary subcortical dysfunction exhibited decreased entropy, similar to those under sedative agents. Conversely, comatose patients with primary cortical dysfunction, such as alpha-coma patients, showed pathologically elevated levels of entropy, exceeding those of wakeful subjects, yet without correlation with cognitive function.

Furthermore, UpToDate reviews on “Stupor and coma in adults” (Young, 2025), “Stupor and coma in children” (Michelson et al, 2025) and “Myxedema coma” (Ross, 2025) do not mention quantitative EEG as a management tool.

Monitoring / Predicting the Response of Treatment in Patients with Major Depressive Disorder

Arıkan et al (2025) noted that qEEG could facilitate the monitoring of treatment progress and the evaluation of therapeutic responses in patients with MDD. These investigators compared the qEEG data of MDD patients and HC, both before and after treatment, to examine the effect of treatment response on neural activity. A total of 72 patients, aged 18 to 60 years, who had not used any psycho-pharmacotherapies for at least 2 weeks, were included in the study. Based on a minimum 50 % reduction in scores on the Hamilton

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Depression Rating Scale (HDRS-17) and Hamilton Anxiety Rating Scale (HARS), Subjects were divided into 2 groups: responders (n = 51) and non-responders (n = 21). qEEG data were recorded before and after treatment. Responders exhibited a significant shift in cortical activity, especially in theta, alpha, and high-beta power, toward patterns resembling those observed in the HC group (improvement range of 15 % to 67 %). In contrast, non-responders showed minimal changes in cortical activity (improvement range of 38 % to 46 %). These findings suggested that while qEEG spectral data reflected significant neural changes in responders, no significant alterations were observed in non-responders. The authors concluded that the use of qEEG spectral analysis to monitor MDD patients provided valuable insights into treatment effectiveness. The distinct patterns of cortical activity observed across most brain regions before treatment, compared to HC, highlighted the potential of qEEG to predict treatment outcomes. Well-designed studies are needed to validate these preliminary findings.

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